

Class 30 Geometry of chaotic attractors

- The 2D system analysis (Gradescope) is due Saturday.
- The weekly project update (Canvas) is due Saturday.
- The 2D system analysis is an individual assignment with **no collaboration**. You may consult with course staff (individually) via office hours or by posting on Ed.

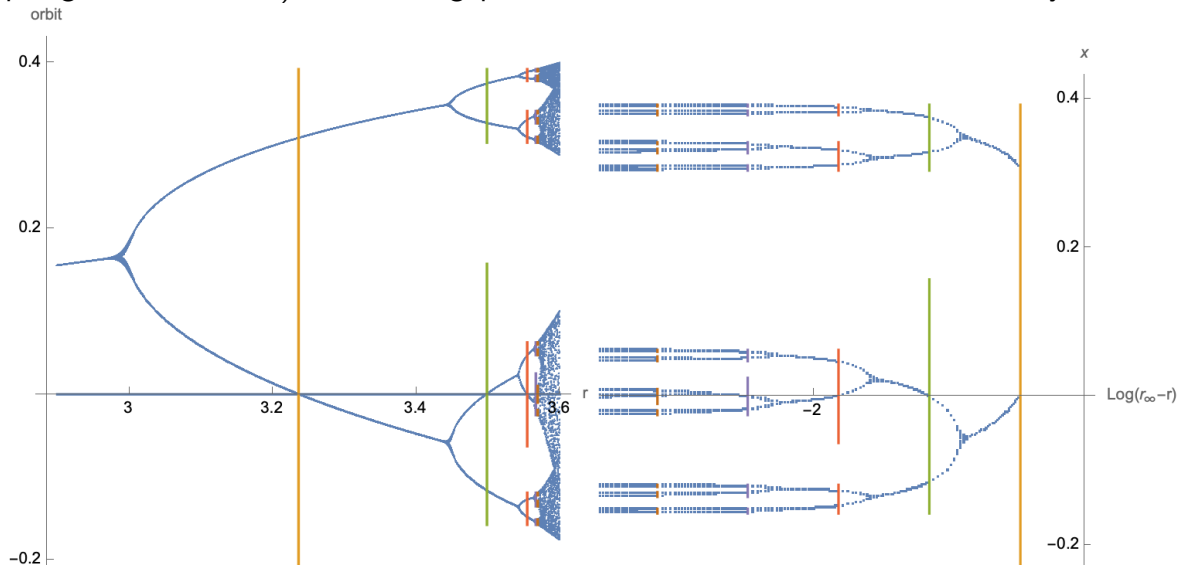
today

We return to a 3D flow.

The Rössler system makes stretching and folding in phase space more visible in a continuous-time system.

We use a return-map-type construction, based on successive local maxima, to connect this flow to the map ideas we have studied, including period doubling and fractal geometry.

The series of period doubling bifurcations in the logistic map leads to a Cantor-like structure (a “topological” Cantor set), where the gaps are various sizes and the set is not strictly self-similar.



On the left is the usual orbit diagram. On the right, I am showing r on a log scale based on the distance to r_∞ , the limiting r value associated with the period-doubling cascade.

Based on this, the limit of the cascade appears to be a Cantor set, meaning that the initial structure of the chaotic attractor appears to be a Cantor set.

Teams 5 and 6: Post photos of your work to the course Google Drive today. Include words, labels, and other short notes that might make those solutions useful to you or your classmates. Find the link in Canvas.

Questions

1. (The Rössler system)

Prof Strogatz describes the Rössler system as trying to capture the stretching and folding from a machine making taffy. The variables do not have specific physical meaning. The system is designed to provide a simple model of stretching and folding in phase space.

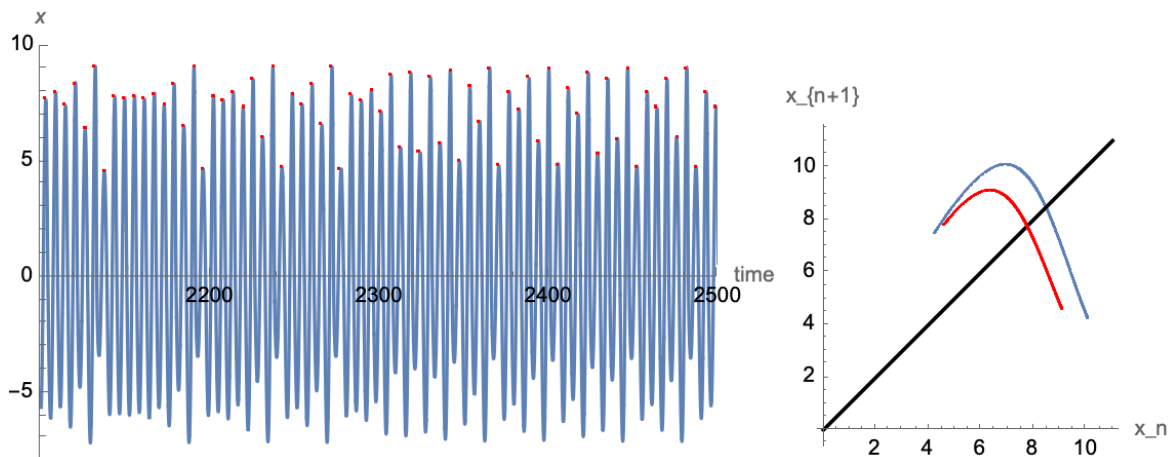
The Rössler system is

$$\begin{aligned}\dot{x} &= -y - z \\ \dot{y} &= x + ay \\ \dot{z} &= b + z(x - c).\end{aligned}$$

This system is slightly simpler than the Lorenz system, with slower volume contraction.

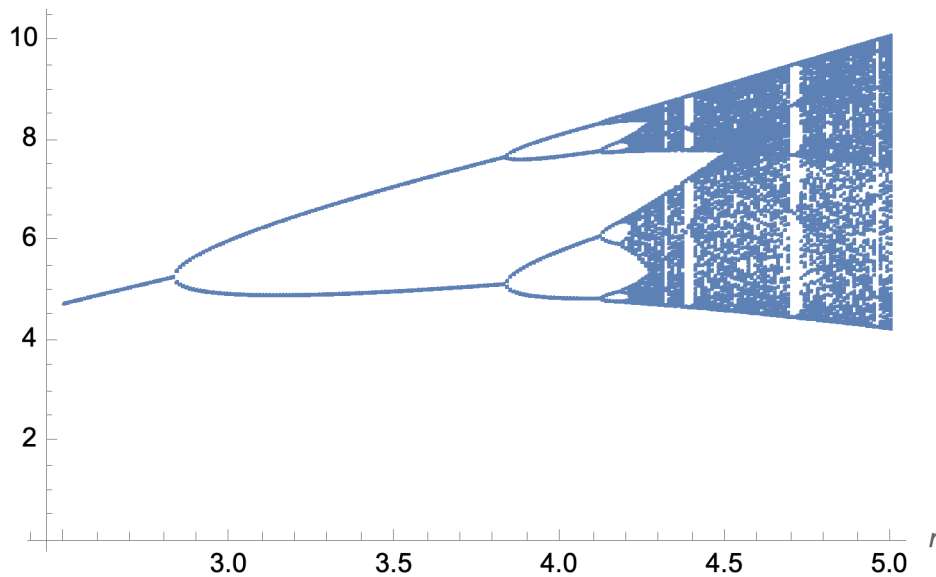
The sketches at the end of the handout, from Abraham and Shaw (1983), illustrate how trajectories in the Rössler system are stretched and folded in phase space.

- (a) Find the nonlinear term(s) in the system above.
 - (b) In the illustrations, imagine starting with a thin band of initial conditions. Where does the band appear to stretch under the flow?
 - (c) Where does that band appear to fold or wrap back?
 - (d) What does this stretching do to small differences in initial conditions?
 - (e) If this stretching and folding repeats many times, what kind of layered structure would you expect to develop? Compare this to the way $S_1, S_2, S_3, \dots$ approach the Cantor set.
2. We reduce the Rössler flow to a discrete sequence by recording successive local maxima of $x(t)$.
- (a) How is this related to a Poincaré map?
 - (b) What information does this keep? What does it discard?
 - (c) On the left is a trajectory with local maxima marked in red. On the right, each maximum is plotted against the next one (for $c = 4.5$ in red and $c = 5$ in blue).
- These plots are not exact maps, since they have some width. Does a one-dimensional map seem like a reasonable model for what you see?



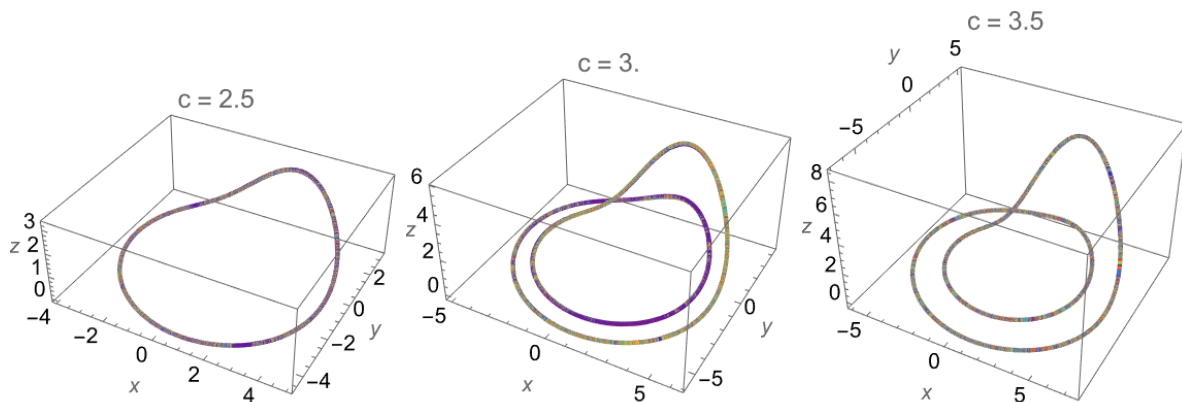
3. (varying a parameter) For each value of c , I compute a long trajectory and plot the local maxima of $x(t)$ against c (using later times in the trajectory).

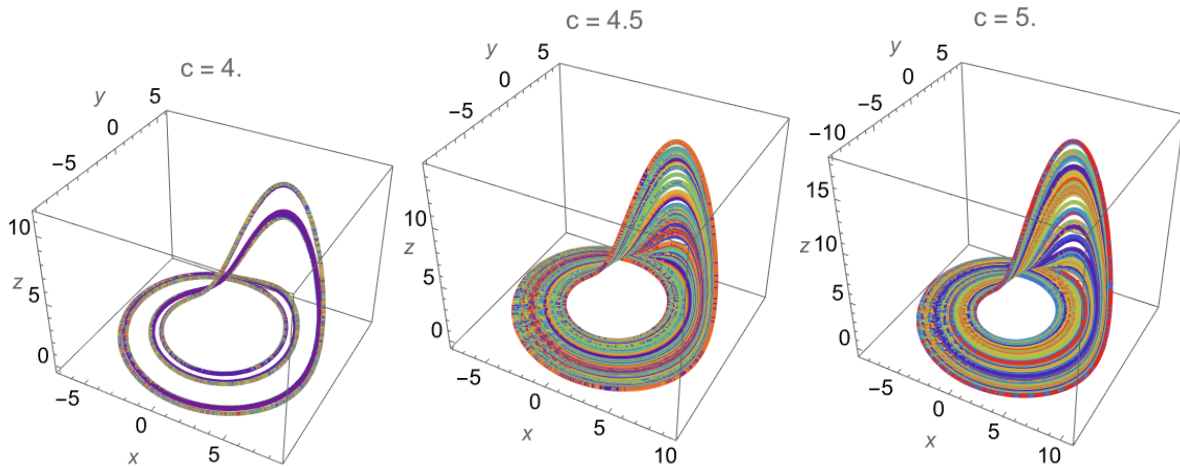
orbit points



- What features of this orbit diagram look similar to the logistic map orbit diagram?
- Where do you see periodic behavior?
- Where do you see period-doubling?
- Where do you see aperiodic or chaotic-looking behavior?

4. Here are some trajectories of the Rössler system at various values of c . You can hopefully see periodic structures from the orbit diagram appearing as trajectories in phase space.





- What periods do you see? Which trajectories seem aperiodic?
- How does the geometry of the trajectory change as the period increases?
- How do these phase-space pictures reflect the structures you saw in the orbit diagram?

Answer:

- just one, in $\dot{z}$
 - this looks kind of like the logistic map (single peak)
 - the orbit diagram is similar to the logistic
 - 1, then 2, then 2, then maybe 4, then the next two look aperiodic
5. How are ideas from the maps phase of our course appearing in the Rössler system?
- You might think about return maps, stretching and folding, period-doubling, orbit diagrams, and fractal structure.*

This attractor also connects back to fractals and fractal dimension.

A chaotic attractor can often be thought of locally as having a smooth sheet-like direction together with a fractured, Cantor-like transverse direction.

We expect a strange attractor in a 3D flow to have dimension between 2 and 3: it is not a full 3D region, but it is more than a smooth 2D surface because of the Cantor-like fine structure.

All semester, we have been studying long-term behavior and how it changes as parameters change.

In chaotic systems, that behavior can be complicated. In the examples we have studied, it is complicated but still has substantial geometric and dynamical structure.

